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CE205
Databases and Information Retrieval

Information Retrieval Concepts and Document Pre-Processing

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Some contents are based on/extracted from Ronan Cummins slides of module information Retrieval

What's done and what's next

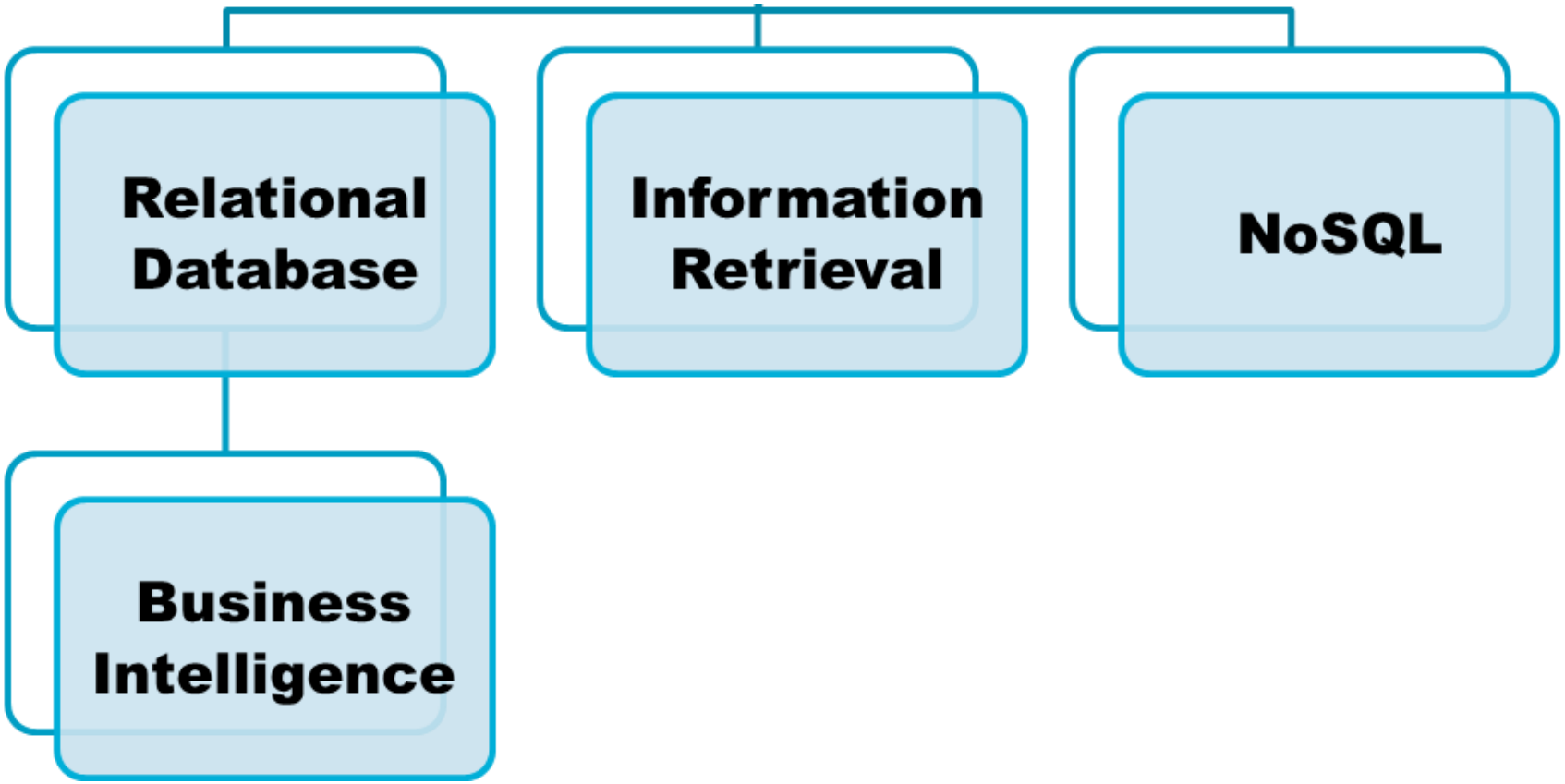
Databases



What we going to learn ????

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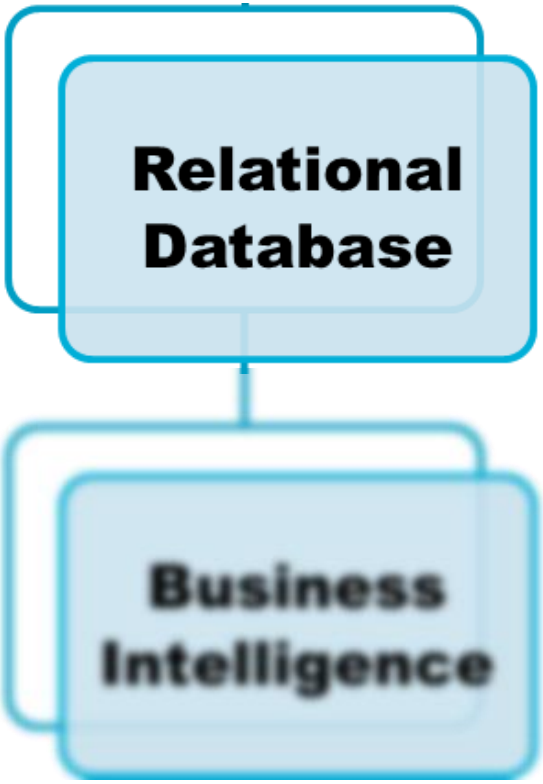
What's done and what's next



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What's done and what's next

Design and Implementation of a Database



- ✓ **Functional Dependency**
- ✓ **Armstrong's Axioms in FDs**
- ✓ **Candidate Key(s) in a Relation (using Closure of Attributes)**
- ✓ **Types of FDs**
- ✓ **Normalisation to 2NF, 3NF, BCNF**
- ✓ **Lossless Join Decomposition**
- ✓ **Discussion on 4NF and 5NF**
- ✓ **Database Security**

What's done and what's next

Design and Implementation of a Database



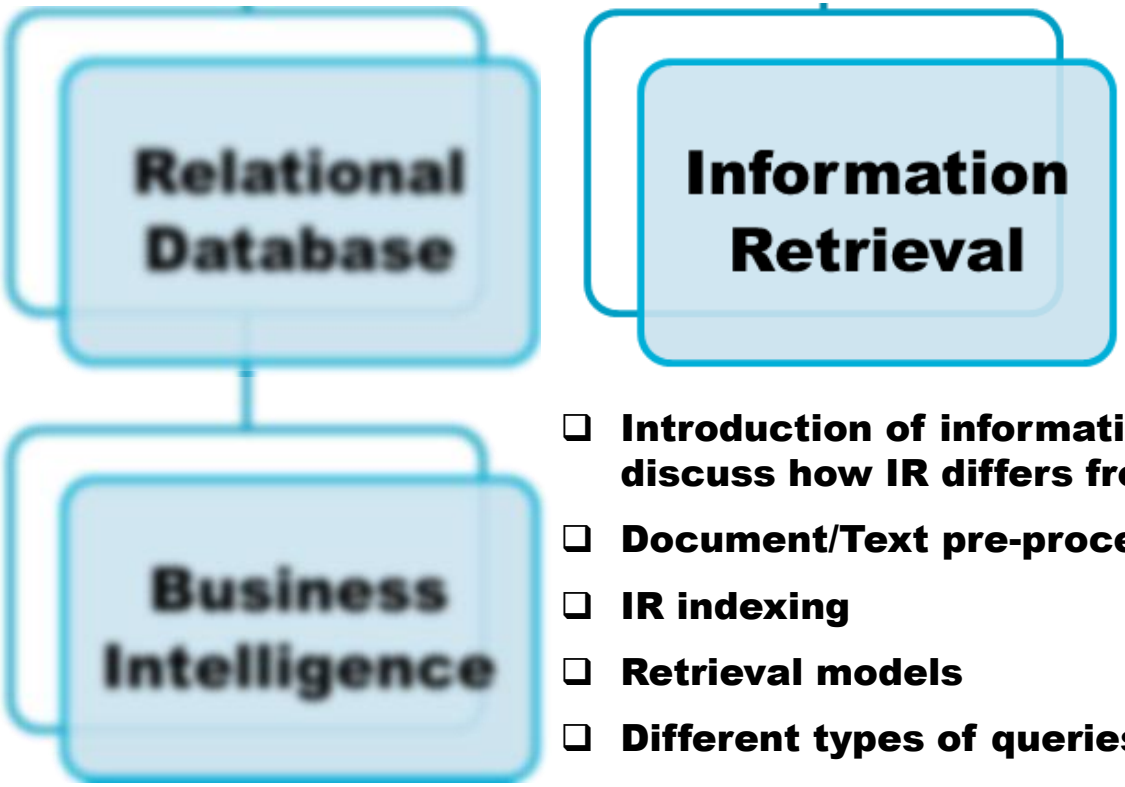
Business Decision Using Databases

- ✓ Functional Dependency
- ✓ Armstrong's Axioms in FDs
- ✓ Candidate keys in a Relation (using Closure of Attributes)
- ✓ Types of FDs
- ✓ Normalisation to 2NF, 3NF, BCNF
- ✓ Lossless Join Decomposition
- ✓ Discussion on 4NF and 5NF
- ✓ Database Security

- ✓ **Why Business Intelligence Systems**
- ✓ **Sources of data in BI Systems**
- ✓ **Data warehouses and Data marts**
- ✓ **Categories of BI Systems**
  - **Reporting Systems**
  - **Data mining applications**

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What's done and what's next



- Introduction of information retrieval (IR) concepts and discuss how IR differs from traditional databases
- Document/Text pre-processing
- IR indexing
- Retrieval models
- Different types of queries in IR systems
- Various evaluation metrics for IR systems

Information Retrieval

- Information retrieval (IR) is **finding material** (usually documents) of an **unstructured nature** (usually text) that satisfies an **information need** from within large **collections** (usually stored on computers).



This field has existed even longer than the database field.

It was originally concerned with retrieval of cataloged information in libraries based on titles, authors, topics, and keywords.

Book and Material (1)

Handbook of Statistics

Chapter 11

Information Retrieval: Concepts, Models, and Systems

by

Venkat N.Gudivada,

Dhana L.Rao,

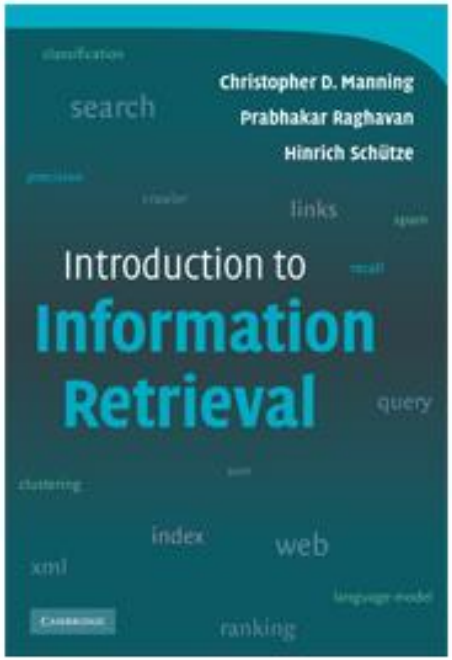
Amogh R.Gudivada

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Book and Material (2)

- Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press. Available online at http://nlp.stanford.edu/IR-book/



- ***Note:*** *This is an advanced book. In a few pages it covered much of the material we will deal with in this course. It is a very good text and it has many references to published papers on specific topics which you can usually download. However, do not worry if you do not understand everything in this book.*

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Useful Resources

- Information Retrieval

https://www.cl.cam.ac.uk/teaching/1516/InfoRtrv/materials.html

Today's Learning Objectives

- What is information retrieval
- Structured, Unstructured, and semi-structured data
- Comparison between relational databases and information retrieval (IR) systems
- Categorization of IR systems
- Modes of interaction with IR systems
- Search engines and web crawling
- Structure of an IR system
- Document Pre-processing (token normalization, Lemmatization, Stemming etc.)